

UNIT - 1. LOADS ON AIRCRAFTS

①

LOAD FACTORS

Aircraft loads are forces & loads applied on a aircraft structural component.

Some Causes of loads on Aircraft Components are

- (a) Air pressure (b) inertia forces
- (c) reactions from ground during landing and take off.
- (d) Arrested landings (e) landing on water, etc.

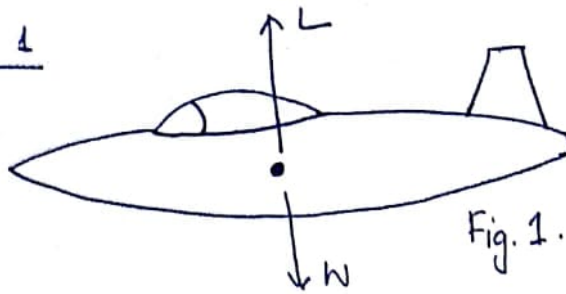
At times there are additional loads that act on the aircraft. Example when an aircraft flies through turbulence or when the aircraft does an maneuver. These additional loads will increase or decrease the net load on the aircraft structure.

The severity of the additional loads depends on the maneuvers or the turbulence, and its magnitude is measured by load factors.

So hence, load factor is a multiplying factor which defines a load in terms of its weight.

We will now look through an illustration to be able to understand what is a load factor.

Case 1



Gust Load

Fig 1 represents the aircraft in straight level flight:

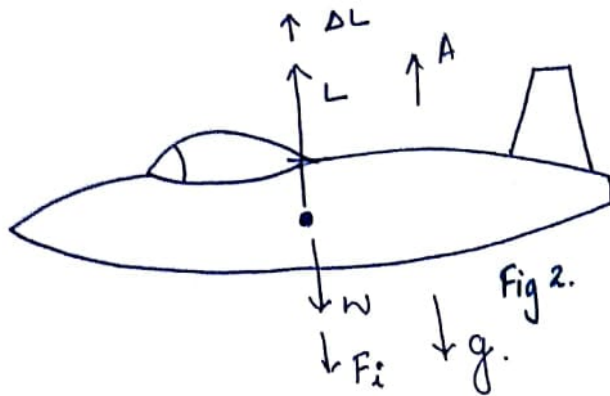


Fig 2 represents the aircraft is subjected to an acceleration normal to its flight path.

Consider the accelerating force to be ΔL ,

Which is an increment in lift, may be caused because of gusts or a maneuver.

This additional lift forces an acceleration A and the equilibrium is provided by the inertia force P .

Hence, we can formulate an equation,

$$\Delta L = F_i = \left(\frac{W}{g} \right) \cdot A \quad \text{or} \quad \frac{\Delta L}{A} = \frac{W}{g} \quad \text{or} \quad \frac{\Delta L}{W} = \frac{A}{g}$$

or

$$\Delta L = \left[\frac{W}{g} \right] A$$

We can reformulate it as,

$$\Delta L + L = W + \frac{W}{g} \cdot A$$

$$\Delta L + L = W \left[1 + \frac{A}{g} \right]$$

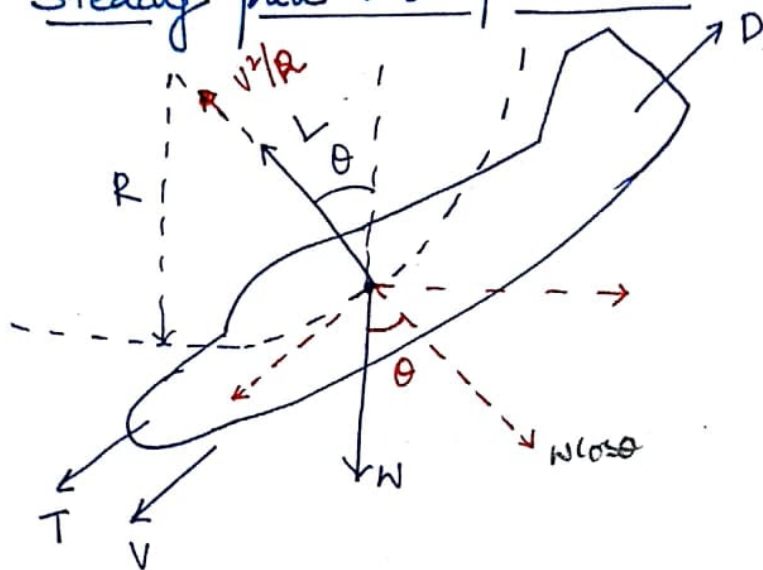
Hence the load factor can be given as,

$$n = 1 + A/g = 1 + \Delta L/W. //$$

Hence, we can say,
 $L = nW$.

(3)

Case 2 Steady pull out from dive.



Let us assume the aircraft has pulled out of a nose dive, and it describes a curve path with radius R . Hence, the centripetal acceleration V^2/R .

where V is the velocity of the aircraft.

We can say lift $L = \frac{WV^2}{gR} + W \cos \theta$. $\left[\frac{W}{g} = m, \frac{V^2}{R} = a \right]$

The quantity $\frac{WV^2}{gR}$ is the magnitude of inertia forces at various points.

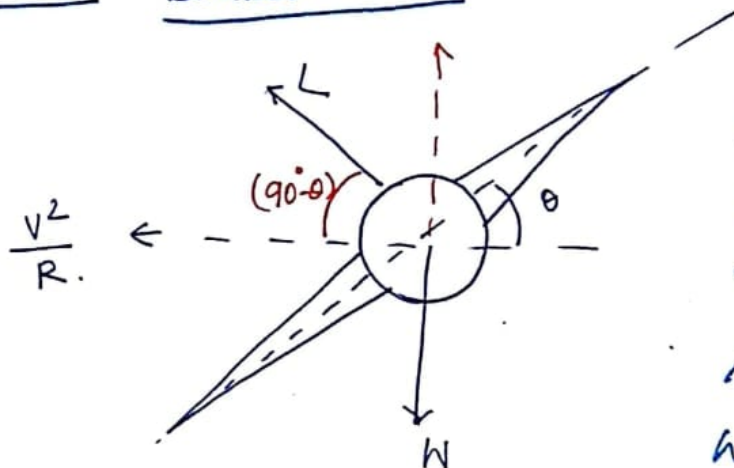
Because we know $L = nW$

$$n = \frac{V^2}{gR} + \cos \theta$$

When $\theta = 0$, at lowest point of the pull out-

$$n = \frac{V^2}{gR} + 1 //$$

Case 3 Banked Turn :-



Assuming the angle of bank to be θ and the radius of the banked turn to be R

We can derive an equation, which will be

$$L \sin \theta = \frac{v^2}{R} \frac{W}{g} \quad \text{but for equilibrium.}$$

$$L \cos \theta = W$$

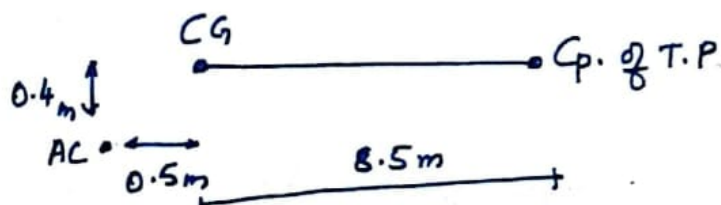
Hence, we get, $\frac{L}{W} = \frac{1}{\cos \theta} = n$

hence $n = \sec \theta$ or $\frac{1}{\cos \theta}$

Example Problem

An Aircraft Weighing 145000 N has Wing Area of 50 m^2 and mean chord of 2.5 m. $C_D = 0.021 + 0.041 C_L^2$ and $dC_L/d\alpha = 4.8$

Allowing downwash, moment about aerodynamic center is $C_m = -0.032$ and Geometric Data is as mentioned below.



During a steady glide with zero thrust at 250 m/s EAS in which $C_L = 0.08$, taking $\rho = 1.223 \text{ kg/m}^3$, Calculate the change in lift when a downward gust is encountered by the aircraft.

(5)

Solution:

$$C_D = 0.021 + 0.04 C_L^2$$

$$\text{Sub } C_L = 0.08$$

$$C_D = 0.021 + 0.04 (0.08)^2$$

$$\text{We get } C_D = 0.0213$$

$$L = W = 145000 \text{ N}$$

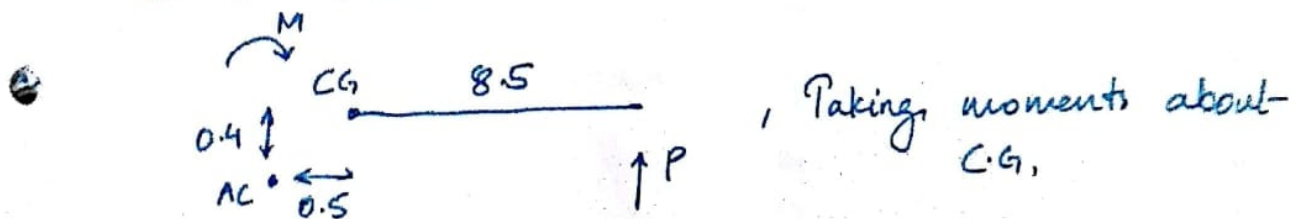
$$\text{from the lift equation } L = \frac{1}{2} \rho V^2 S C_L$$

$$\text{or } L = \frac{1}{2} \rho V^2 S \frac{dC_L}{d\alpha} \alpha_w, \text{ where } \alpha_w \text{ is the Angle of attack of the wing.}$$

$$L = \frac{1}{2} \rho V^2 S \frac{dC_L}{d\alpha} \alpha_w$$

$$145000 = \frac{1}{2} \times 1.223 \times 250^2 \times 50 \times 4.8 \alpha_w$$

$$\alpha_w = 0.0158 \text{ rad or } 0.91^\circ$$



$$L \times 0.5 - D \times 0.4 + M = P \times 8.5$$

$$0.5(L + P) - \frac{1}{2} \rho V^2 S C_D + \frac{1}{2} \rho V^2 S \bar{c} C_m = 8.5P$$

Substituting and solving, we get,

$$P = -10740 \text{ N}$$

$$\text{Total Lift} = , L = W - P = 145000 + 10740 = \underline{\underline{155740 \text{ N}}}$$

V-N diagrams:-

The operating flight strength limitation of an airplane are presented in the form of a V-N diagram. A typical V-N diagram is shown in the figure below.

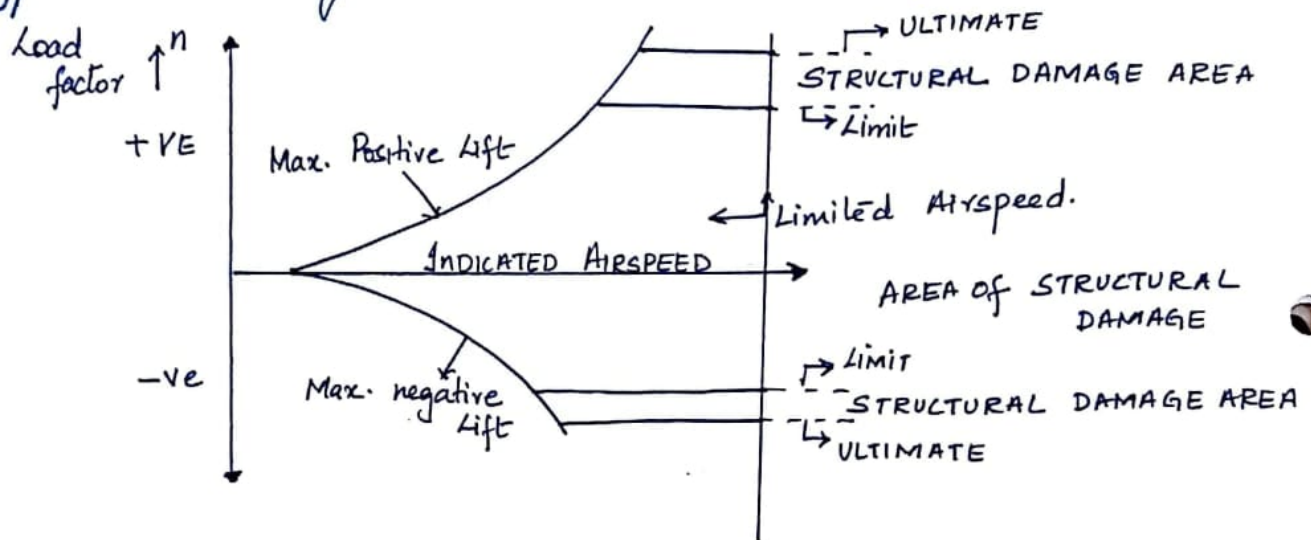


Fig. Showing a typical V-N diagram.

The horizontal scale is the airspeed, and Vertical scale is the load factor.

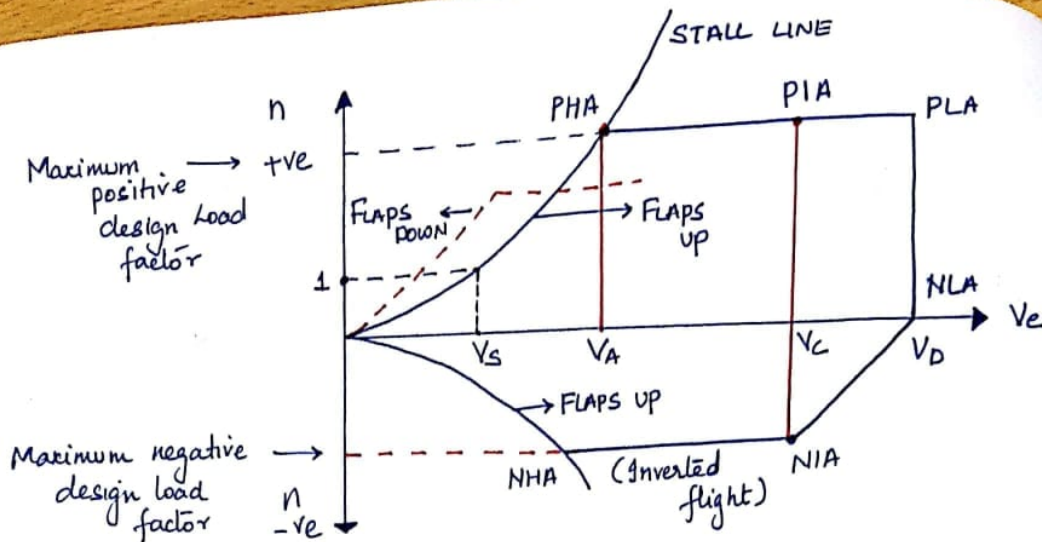
The airplane strength is dependent on

- (a) Gross weight (b) Configuration of the aircraft
- (c) Symmetry of loading (d) Altitude.

A change in any of these parameters can cause important changes in operating limits.

A typical V-N diagram is shown for Maneuver's. in the fig below.

V-N diagram



- PHA → Positive High Angle of Attack
- PIA → Positive Intermediate Angle of Attack
- PLA → Positive Low Angle of Attack
- NHA → Negative High Angle of Attack
- NIA → Negative Intermediate Angle of Attack
- NLA → Negative low Angle of Attack.

Stall line indicates maximum load factor that can be obtained because of the inability of the aircraft to produce lift at a given airspeed.

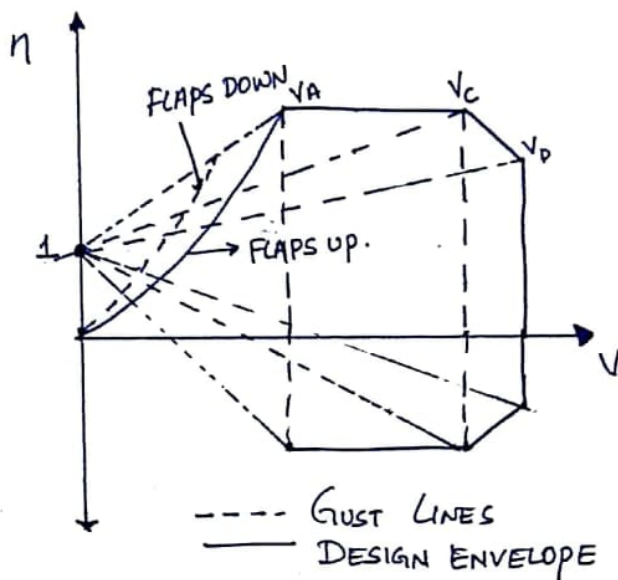
V_s indicates the velocity at which the aircraft can fly with a load factor 1.

V_A is the velocity it can fly with a +ve high AOA within the flight envelope without structural damage.

Similarly V_c and V_d correspond to intermediate and low angle of attack.

When the flap's are deployed the maximum design load factor reduces as indicated by the red line in the above fig.

V-N diagram for Gust Envelope:-



Gust lines radiate from $n=1$ at $V=0$.

Constructed Using formulae

$$n = 1 \pm \frac{k_g U_{de} V_e C_{L\alpha}}{498 [W/s]}$$

$k_g \rightarrow$ alleviation factor.

$U_{de} \rightarrow$ Gust Velocity.

$V_e \rightarrow$ Equivalent Airspeed.

Here, we can see that when there is a gust-load the load factor decreases and is within the flight envelope.

Design gust intensity reduces as the Velocity increases. Hence, as the turbulence increases the airplane shown fly at reduced speeds.

WING DESIGN LOADS

Wing loads consist of Shear, bending and Torsion which result from inertial loads, air pressures, etc.

Other flight loads include, flight control surface deflections and wing loads must also be designed for landing and Taxi conditions.

Airload Spanwise distribution:-

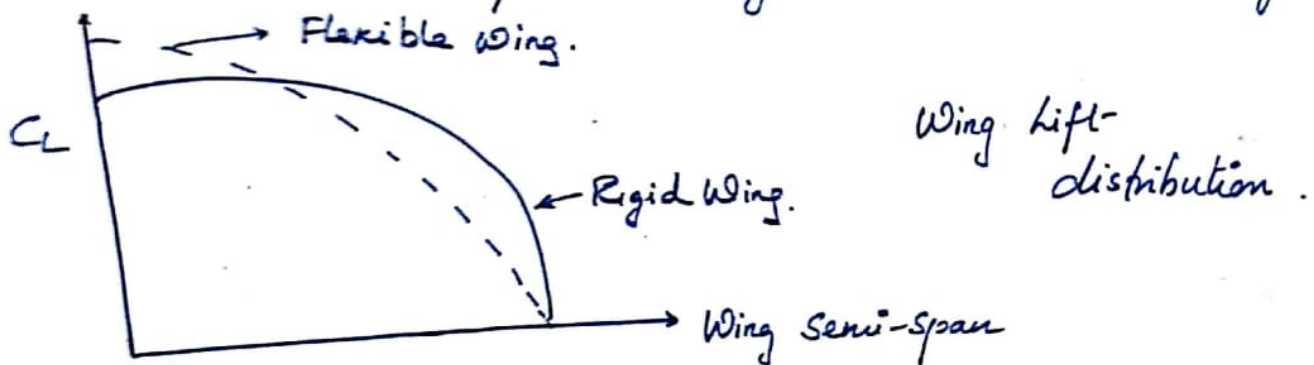
Air loading on the wing consists of two parts

- (a) Basic loading.
- (b) Additional loading.

The additional air loading is caused by, the angle of attack.

On normal aspect ratio wings, with Aspect ratio > 3 lift varies directly with the angle of attack.

The basic loads are lift and drag that act on the wing.

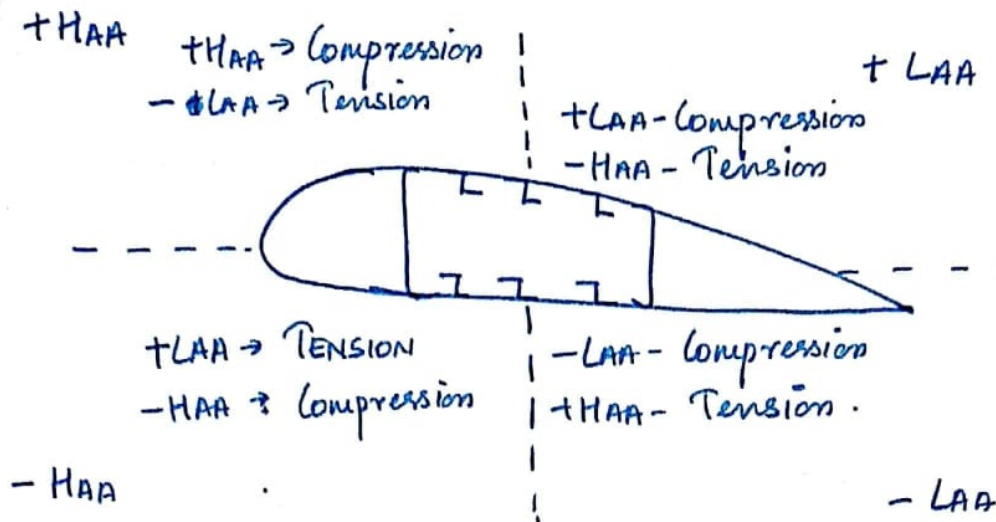


Fuselage, Nacelle and Wing stores effect on Wing loads:-

Stores/Nacelle located at the wing cause protuberances and if a tip tank is present, it will effectively increase the aspect ratio.

And a radome present at the wing ^{root} tip will interfere with the lift, causing the airplane to fly at a higher angle of attack, causing higher loads at the outboard

For the stress analysis of the conventional wing, it will be necessary to investigate each cross section for 4 conditions as shown in the figure.



The four conditions are

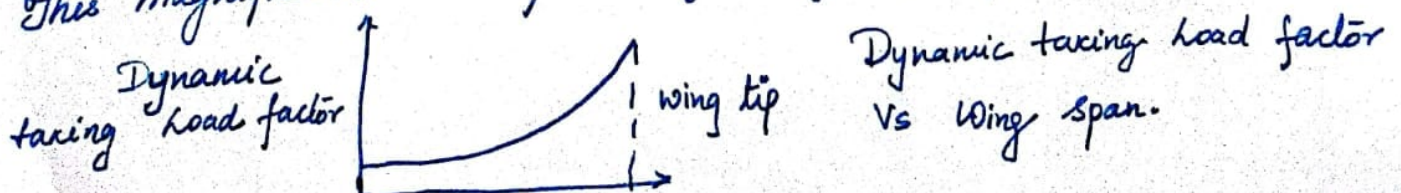
- +HAA $\rightarrow$ positive high angle of attack
- HAA $\rightarrow$ Negative " " " "
- LAA $\rightarrow$ " LOW " " "
- +LAA $\rightarrow$ Positive " " " "

LANDING AND TAXI Effect on Wing Loads:-

If the gear is located on the wings, then there will be large loads applied on the wing during landing.

Taxiing must be accomplished at maximum take off weights, when the load is maximum, the wing is full of fuel and the wing stores have been installed.

Rough runways will cause an increase in the load factors. Whipping of the wing during landing and taxi causes an magnification of the load factors experienced at the wing tip. This magnification is a function of weight & stiffness.

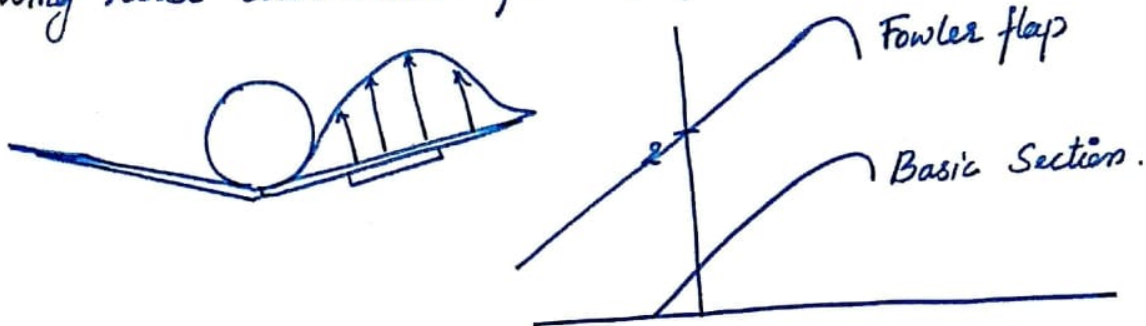


Wing Control Surfaces:-

Tail loads

Airplanes equipped with wing flaps, slats, ailerons, spoiler's, dive brakes, and other high lift devices.

① Wing load distribution from flap extension



② Wing load distribution due to deflected aileron's.



Wing Weight:-

To determine the design loadings, The weight distribution must be taken into account.

- ① Basic wing structure
- ② Power plant
- ③ Landing gear (if mounted on the wing).
- ④ Removable stores, (fuel tanks etc).

Tail loads / Empennage loads:-

Tail loads include both the loads acting on the aircraft's horizontal stabilizer and vertical stabilizer. The load on the horizontal stabilizer is due to the following conditions.

- (a) C_L due to AOA of H.T
- (b) C_L increment due to elevator deflection.
- (c) Hinge moments.
- (d) Positive and negative maneuvers.
- (e) Fin gust (in T-Tail).

Vertical tail loads,

The following reasons could add additional load on the vertical tail.

- (a) Yaw initiation or instantaneous rudder movement.

Rudder maneuvers are of 2 types:-

- (1) Rapid movement of the rudder with pedal force.
- (2) Oscillating rudder motion.

- When the rudder is deflected, the effect on the vertical ~~load~~ tail is due to the rudder deflection.

Figures show the inertia loads, hinge loads and rudder loads., also the figures show the

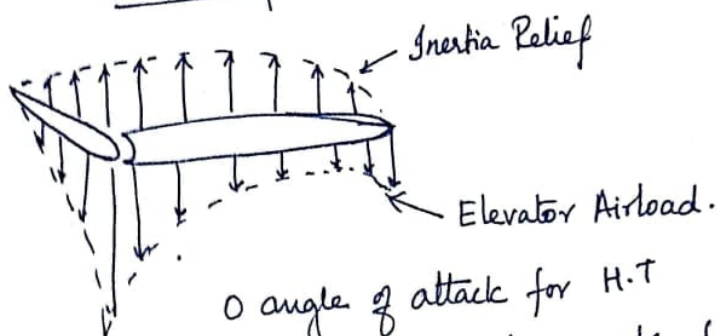
- loads due to elevator deflection in horizontal tails.

Pitch up Maneuver Horizontal tail Critical points:

Deflec

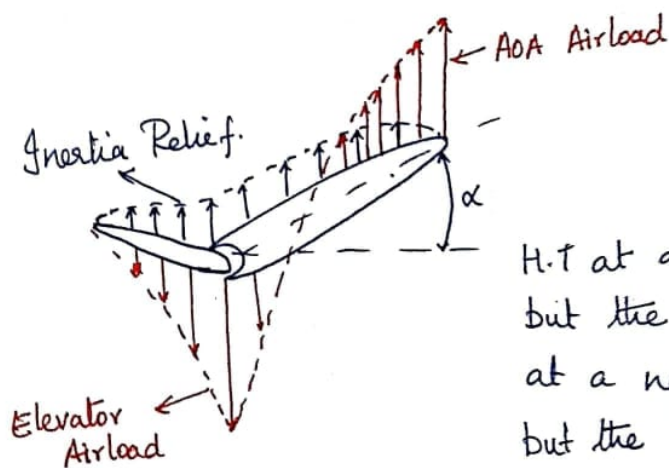
Case I

Case I



0 angle of attack for H.T.
 Elevator deflected upwards for pitch up maneuver.

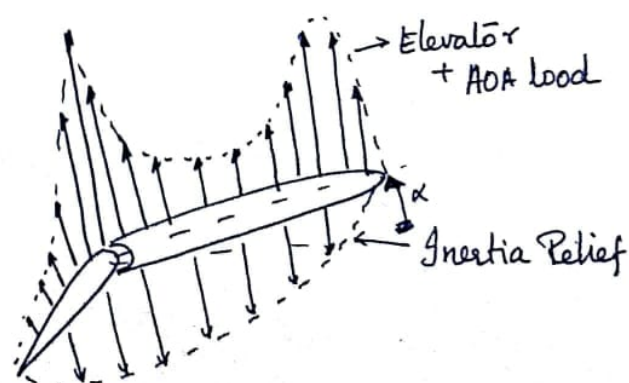
Case II



High torsional loads on the Horizontal Tail plane.

H.T. at a AOA of ' α ' but the elevator deflected at a negative AOA but the net effect is a pitch up maneuver.

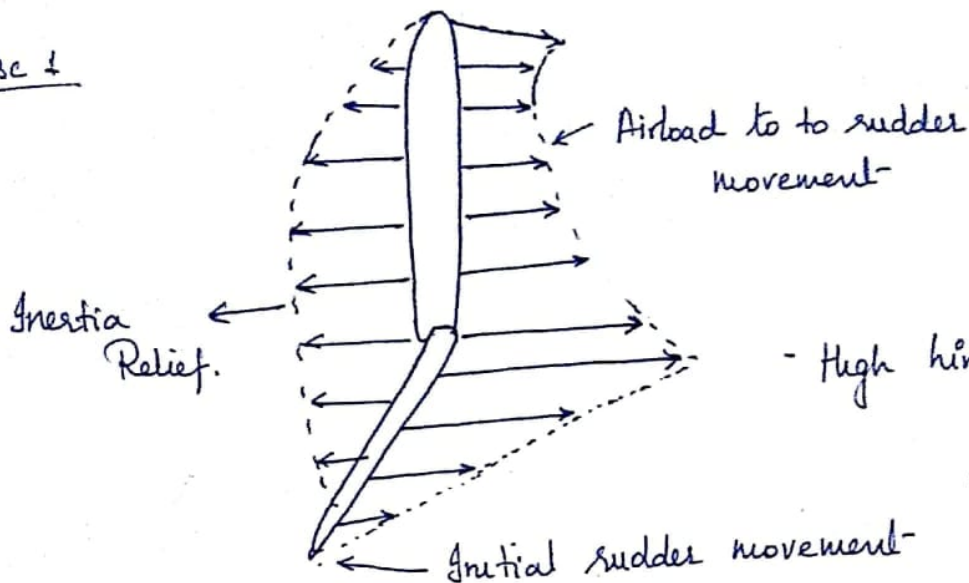
Case III



H.T. is at an AOA of ' α ' and the elevator is deflected at a positive AOA causing a pitch down maneuver.

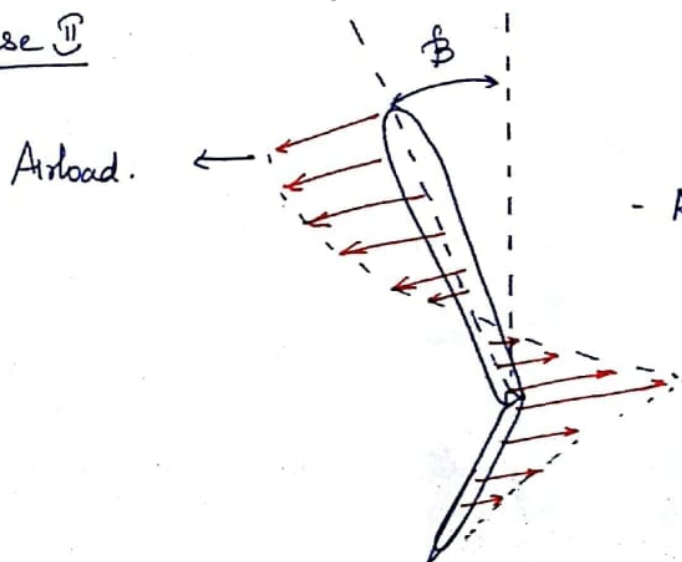
Deflection of the rudder causing loads on the Vertical tail

Case I



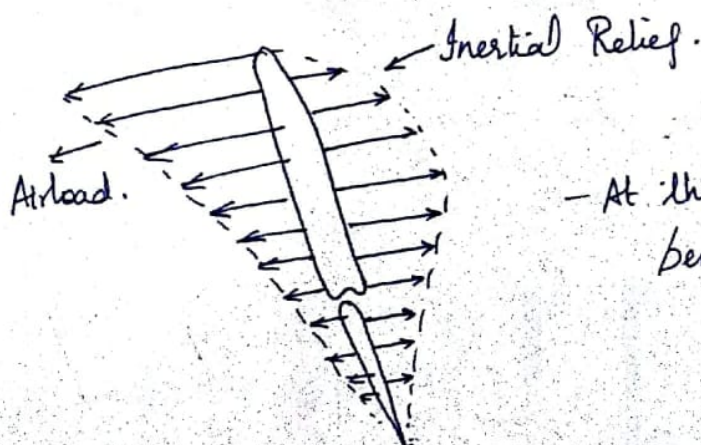
- High hinge loads and Rudder loads are generated.

Case II



- At this point, high torsional loads are generated, on the Vertical tail.

Case III

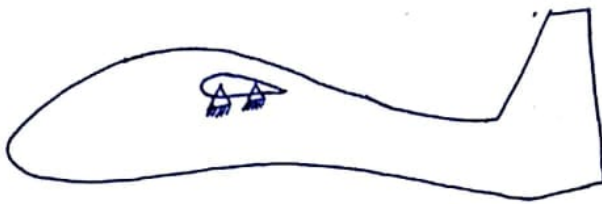


- At this moment maximum bending moment occurs.

Fuselage loads :-

Types of loads acting on the fuselage:-

- (a) Vertical bending
- (b) Horizontal bending
- (c) Torsion.



Considering the fuselage as a statically determinate beam fixed at the wing-fuselage joints.

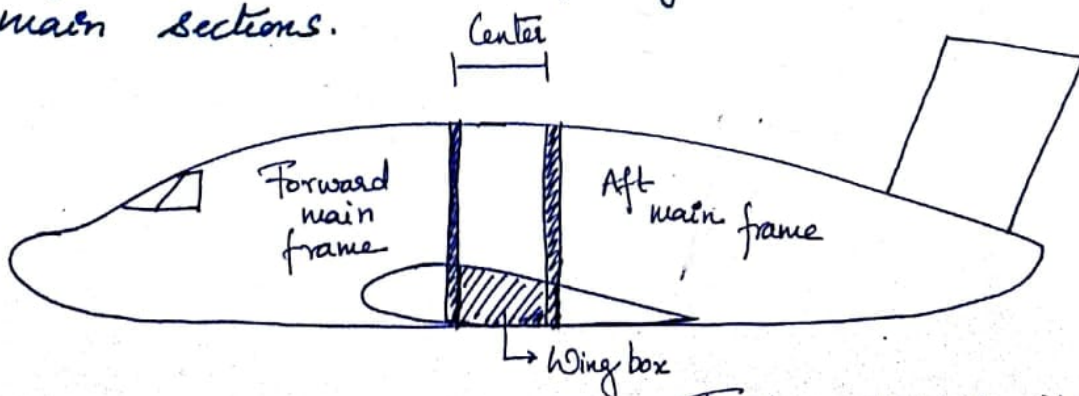
Distribution of Weight is an important parameter in determining loads acting on the fuselage.

Loads on the fuselage can be due to:-

- ① Maneuvers ② Landings ③ Ground handling.

- Tail loads contribute to bending of the aft portion of the fuselage.
- Di-symmetry of the tail loads aft-fuselage torsion.

For a typical airframe, the fuselage is divided into 3 main sections.



- Internal pressures and external are some of the major loads acting on the fuselage.

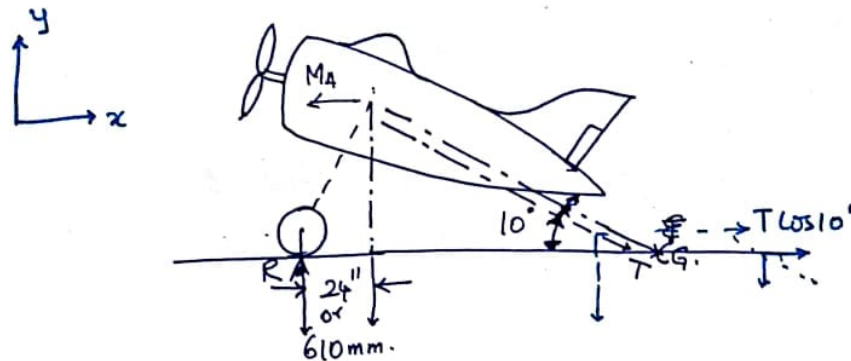
Fuselage divided into three sections.

Sample Problem:-

(16)

An Aircraft landing on the runway of a Carrier ship is being arrested by a cable pull T on the airplane arresting hook. If the aircraft weighs, 55,000 Newtons, and a constant acceleration of $3.5g$ is given. Find hook pull T , the wheel reaction R and distance ' d ' between the line of action of the hook pull and airplane C.G. If the landing velocity is 30 m/s what is the stopping distance?.

Sol:-



$$M_a = \frac{W}{g} \cdot a = 55,000 \times 3.5 \frac{g}{g} = 192,500 \text{ Newtons. (Inertia acts opposite to acceleration).}$$

from static equilibrium, Unknown forces T & R can be solved.

$$\sum F_x = -192,500 + T \cos 10^\circ = 0$$

$$T = 195,469.62 \text{ N.}$$

$$\sum F_y = -55,000 + R - 195,469.62 \sin 10^\circ = 0$$

$$\Rightarrow R = 88,942.83 \text{ N.}$$

To find distance ' d '. $\sum M_{CG} = R \times 610 - 195,469.62 \times d = 0$

$$d = 277.56 \text{ mm.}$$

Stopping distance :-

$$v^2 - 0^2 = 2as$$

$$30^2 = 2(3.5g) \cdot s$$

$$s = 13.10 \text{ m}$$

Propulsion loads :-

These loads are usually referred to as, Engine loads.

Major Considerations are thrust loads, aerodynamic loads, inertia loads and gyroscopic loads.

Thrust :-

Thrust acts on the aircraft components during take off (Maximum take off velocity), reverse thrust, maximum climb etc. Thrust generally produces high load factors on engines.

Airloads

The inlet portion of the Pod is highly loaded portion of the structure. As the airflow is straightened in this region of the structure, the loads are very high at this region.

→ ~~Dyn~~ Landing, Taxi, and Gust are conditions that often produce largest Engine load factors.

→ Gyroscopic loads or a gyroscopic moment is developed when the rotating parts of the propulsion system are rotated about either the pitch or yaw axis.

Moments are given as,

$$M_z = -I \dot{\theta} \dot{\psi}$$

$$M_y = I \dot{\theta} \dot{\psi}$$

I - polar moment of inertia

N - Engine Angular Velocity

$\dot{\theta}, \dot{\psi}$ = pitch & Yaw Velocity

the fuselage.

Landing Gear loads:-

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The ~~only~~ components of the aircraft landing gear are categorized as safe-life, since a completely fail-safe design is not possible because crack detection is more complicated than it is for other aircraft.

Due to runway roughness and maneuver's (ground operations, take off and landing) repeated loads in the x -, y -, and z - directions are acting on the landing gear simultaneously.

It has been documented that 82 percent of all fatigue failures on landing gear are caused by taxing.

A few loading conditions are mentioned below, which act on the aircraft landing gear.

- (a) Taxiing (c) Braking
- (b) Turning (d) Push back & Towing.

- It is apparent that the landing loads are dependent on a number of variables. among which Sink rate is a major factor. (5-10 ft/s for commercial jets).

- The energy absorption during landing is important, the ability of the landing gear to absorb this energy governs the value of the landing load factor.

Landing conditions also determine the increase in load factor on the aircrafts landing gear. Mentioned below are different landing conditions, for an airplane with tri-cycle gear.

- ① Three point - level landing
- ② Two point - level landing.
- ③ Tail down landing
- ④ One wheel landing
- ⑤ Drift landing.

Landing gear loads associated with ground maneuvers.

- ① Braked roll - 3 wheels
- ② Braked roll - 2 wheels
- ③ Unsymmetrical Braking
- ④ Reverse braking.

Miscellaneous Loads:

The following is a partial list of types of miscellaneous loads.

- (a) Ground handling.
- (b) Control Surface loads
- (c) Jacking loads

The above mentioned are a few miscellaneous loads acting on the aircraft.

$V-N$
limit load:-

V-N diagram Calculations:-

Limit load:- Maximum load the aircraft is expected to experience in normal operations.

Proof load:- Product of limit load and proof factor

Ultimate load:- Product of limit load and Ultimate factor.

$$\text{Load factor} = n = L/W = \frac{1/2 \rho_{\infty} V_0^2 C_{L_{\max}}}{W/S}$$

We can see that load factor 'n' varies quadratically with Velocity 'V'. (Only upto a certain limit)

This Velocity is determined by conditions which are determined Aerodynamically and Structurally. which are $C_{L_{\max}}$ and $V_{\max}$. This Velocity is called 'corner Velocity'.

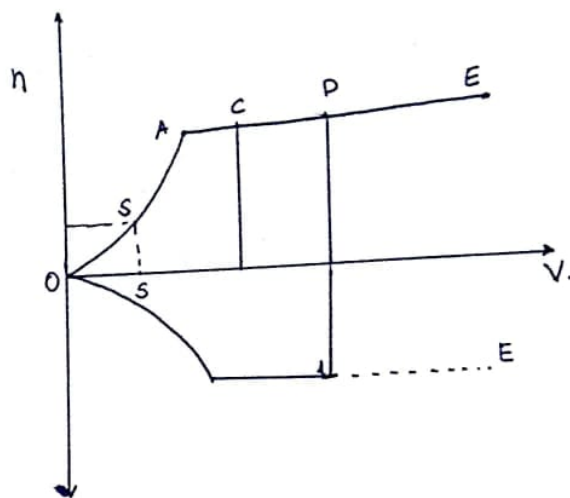
Given by: $V_A = V_{\text{corner}} = \sqrt{\frac{2 n_{\max} \cdot W}{\rho_{\infty} C_{L_{\max}} S}}$

V-N diagram precisely gives the structural and aerodynamic boundaries for a particular flight condition.

The main Velocities that are plotted in the V-N diagram are.

- ① 1-G. Stall Velocity.
- ② Design Maneuvering Velocity.
- ③ Design cruise Velocity
- ④ Design dive Velocity

Appendix for V-n diagram:-



Curve OA is given by.

$$q_{max} = \frac{1}{2} \frac{\rho_{\infty} V_{\infty}^2 C_{Lmax}}{[W/S]} \quad \text{--- (1)}$$

Curve AC is given by.

$$V_C = \sqrt{\frac{\left[\frac{I}{W} \times \frac{W}{S} \right] - \sqrt{\frac{W}{S} \times \left(\frac{I}{W} \right)_{max}^2 - (4K C_{D,0})}}{\rho C_{D,0}}} \quad \text{--- (2)}$$

Gust load formulae is given by.

$$\Delta n_{max} = \frac{m \text{ to } S V_e U_{de} \text{ Kg.}}{2W} \quad \text{--- (3)}$$

$m \rightarrow$ Wing Lift-Curve slope

$\rho \rightarrow$ Density at sea level.

$S \rightarrow$ Wing area

$V_e \rightarrow$ EAS (Equivalent air speed)

$U_{de} \rightarrow$ Gust Velocity effective.

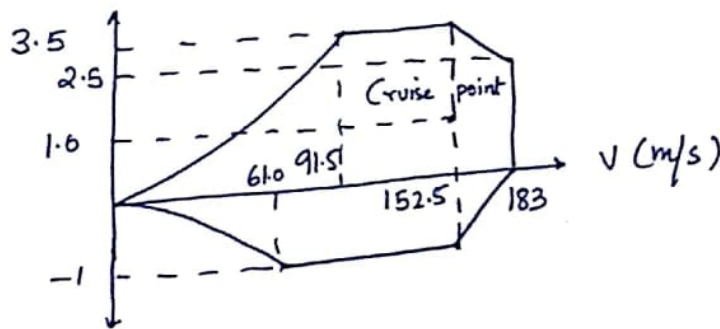
$W \rightarrow$ Airplane weight

$K \rightarrow$ Alteration factor.

Ques 1:-

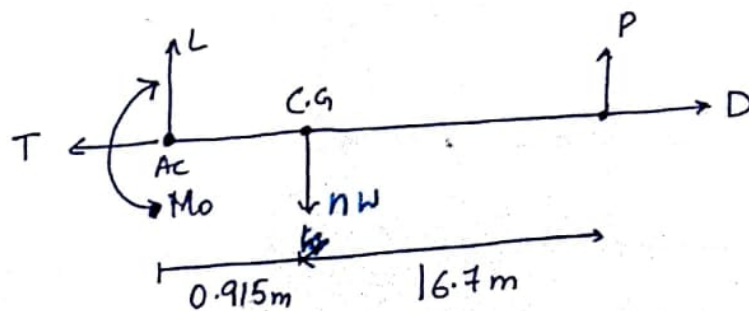
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Fig shows the flight-Envelope at Sea level for a aircraft with $S = 27.5 \text{ m}$. Avg. Wing Chord = 9.30 m & total weight = 196000 N . The Aerodynamic Center is 0.915 m forward of the C.G. and Center of lift for tail Unit is 16.7 m aft of C.G. Pitching moment Coefficient is $C_{m0} = -0.0638$



For steady cruising flight- fuselage bending moment at C.G. is $60,000 \text{ N-m}$. Calculate maximum value of this bending moment- for given flight- Envelope. (Aerodynamic loading on fuselage can be neglected.). Cruise Velocity is 152.5 m/s .

Sol:-



Moments about C.G

$$0.915 \cdot L + M_0 = 16.7 P$$

For vertical equilibrium

$$L + P = nw$$

$$\sum F_y = 0$$

$$L + P = nw$$

$$L = nw - P$$

$$M_0 = \frac{1}{2} \rho V^2 S \bar{C}_{M_0} \quad 9.30$$

$$= \frac{1}{2} \times 1.225 \times 27.5 \times 0.006 \times 0.0638 V^2$$

$$M_0 = -9.98 V^2$$

$$0.915 (nW - P) + M_0 = 16.7 P.$$

$$0.915 nW - 9.98 V^2 = 15.78 P$$

$$0.0579 nW - 0.632 V^2 = P. \quad \text{--- (2)}$$

In Cruise Conditions.

$$n=1 \quad \text{and} \quad V = 152.5 \text{ m/s (Calculated from (2))}$$

$$P = -3349.55 \text{ N.}$$

Bending Moment in the fuselage at the C.G. is given by.

$$M_{CG} = n \cdot M_{lev.ftt} - 16.7 P.$$

When $n=1$

$$600000 = n M_{lev.ftt} + 16.7 (3349.55)$$

$$M_{lev.ftt} = 544062.51 \text{ N-m.}$$

To find Load factor.

Rewriting equation.

$$M_{CG} = n \cdot 544062.51 - 16.7 (0.0579 nW - 0.632 V^2)$$

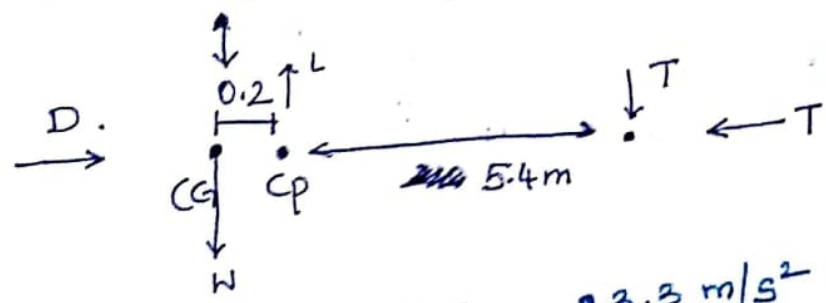
Rearranging.

$$M_{CG} = 379789n + 9.47 V^2 \quad \text{--- (3)}$$

From (3) find max M_{CG} at different n and V .

Ques:- An aircraft weighing 6500 kg's is flying horizontally at a velocity of ~~250~~ 250 m/s when the pilot does a pitch up maneuver into a curved path having a radius of curvature of 750 m. If the engine thrust and drag are equal and opposite.

- Calculate:-
- (a) Acceleration of the airplane
 - (b) Wing lift and Tail forces.
 - (c) Aircraft load factor.



Sol

$$\text{Acceleration} = \frac{v^2}{r} = \frac{250^2}{750} = 83.3 \text{ m/s}^2$$

the 'g' force felt will be $83.3 \text{ m/s}^2 / 9.81 \text{ m/s}^2 = 8.54 \text{ g}$.

Force = $M \cdot a$

$$= \frac{6500}{g} \times 8.54g = 55250 \text{ kg's}$$

Taking moments about CP. (because we do not know L)

$$-(6500 + 55250)0.2 + T \times 5.4 = 0$$

$$T = 2287.03 \text{ kg's}$$

$$\sum F_y = 0 \quad L - W - T - F = 0$$

$$L - 6500 - 2287.03 - 55250 \text{ kg} = 0$$

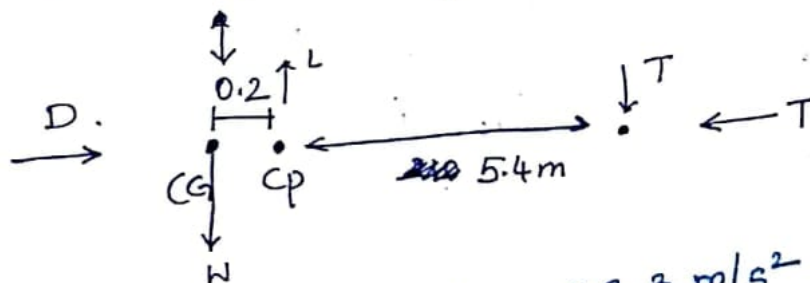
$$L = 64037 \text{ kg's}$$

Hence load factor $L/W = \frac{64037 - 2287.03}{6500} = 9.5$

$n = 9.5$

Ques:- An aircraft weighing 6500 kg's is flying horizontally at a velocity of ~~250~~ 250 m/s when the pilot does a pitch up maneuver into a curved path having a radius of curvature of 750 m. If the engine thrust and drag are equal and opposite.

- Calculate:-
- Acceleration of the airplane
 - Wing lift and Tail forces.
 - Aircraft load factor.



Sol

$$\text{Acceleration} = \frac{v^2}{r} = \frac{250^2}{750} = 83.3 \text{ m/s}^2$$

the 'g' force felt will be $83.3 \text{ m/s}^2 / 9.81 \text{ m/s}^2 = 8.54 \text{ g}$.

Force = $M \cdot a$

$$= \frac{6500}{g} \times 8.54g = 55250 \text{ kg's}$$

Taking moments about C.P. (because we do not know L)

$$-(6500 + 55250)0.2 + T \times 5.4 = 0$$

$$T = 2287.03 \text{ kg's}$$

$$\sum F_y = 0 \quad L - W - T - F = 0$$

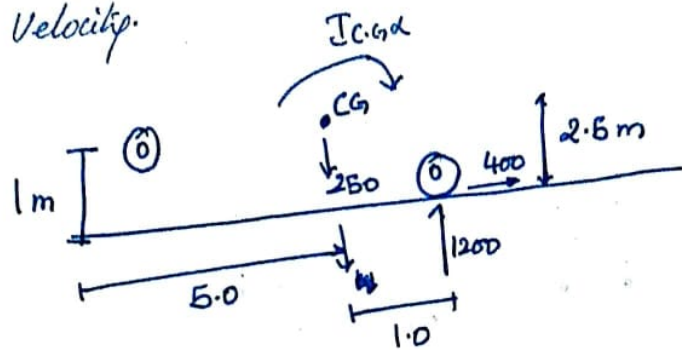
$$L - 6500 - 2287.03 - 55250 \text{ kg} = 0$$

$$L = 64037 \text{ kg's}$$

Hence load factor $L/W = \frac{64037 - 2287.03}{6500} = 9.5$

$n = 9.5$

Problem 1: An aircraft weighing, 250 kN has a tricycle Undercarriage lands with a Vertical Velocity of 8.5 m/s such that the vertical and horizontal reactions are 1200 kN and 400 kN respectively. At this moment, the nose wheel is 1.0 m from the ground as shown. If the moment of inertia of the aircraft about the C.G. is $5.65 \times 10^8 \text{ N-s}^2\text{-mm}$, Determine the inertia forces on the aircraft the time taken for the vertical velocity to become zero and its angular velocity.



Sol: $\sum F_x = 0$, $M_{ax} - 400 = 0$, $M_{ax} = 400 \text{ kN}$, $a_x = 400/M$
 $\sum F_y = 0$, $M_{ay} + 250 = 1200$, $M_{ay} = 950 \text{ kN}$, $a_y = 950/250 = 3.8g$.

Taking Moments about C.G.,

$$I_{C.G.} \alpha = 1200 \times 1.0 + 400 \times 2.5 = 0$$

$$I_{C.G.} \alpha = 2200 \text{ kN-m}$$

Finding Angular Velocity. $\frac{I_{C.G.} \alpha}{I_{C.G.}} = \alpha = 3.9 \text{ rad/s}^2$

$V = u + at$, $V = 0$, when a is deceleration

$$0 = 3.7 - 3.8(9.81)t$$

$$t = 0.099 \text{ sec's}$$

Similarly,

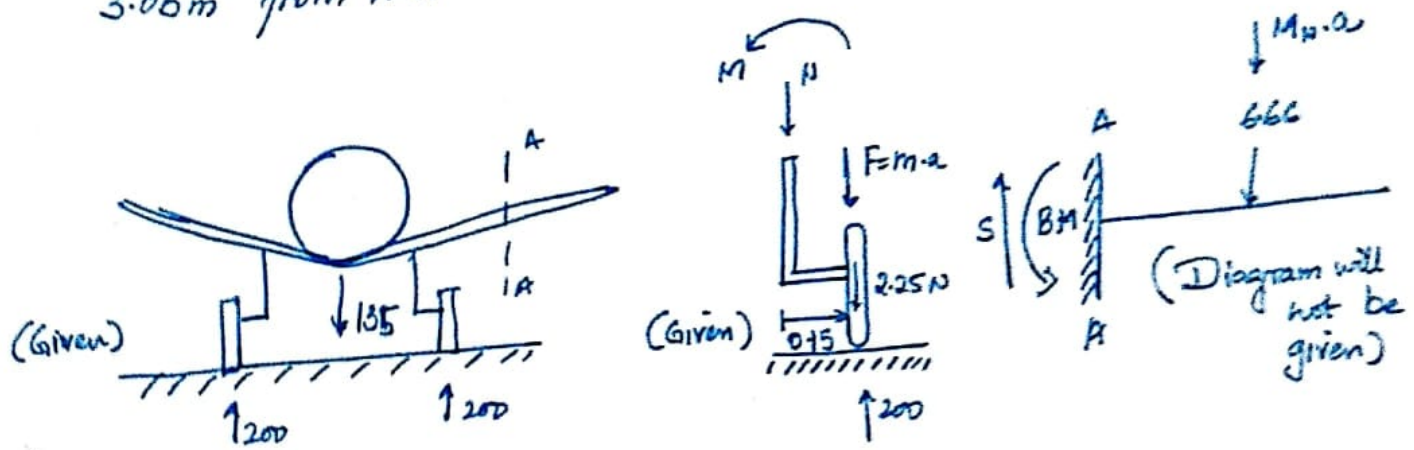
$$\omega = \omega_0 + at$$

$$\omega = 0 + 3.9 \times 0.099$$

$$\omega = 0.39 \text{ rad/s}$$

Q. aircraft shown weighs 135 kN and has landed such that at the instant of impact the ground reaction on each main undercarriage wheel is 200 kN and its vertical velocity is 3.6 m/s. If each undercarriage wheel weighs 2.25 kN and attached to an oleo strut, Calculate.

- Axial and bending moment in the strut, the strut may be assumed to be vertical.
- Determine the shortening of the strut, when the vertical velocity of the aircraft is zero.
- Calculate shear force and bending moment in the wing section A-A if the outboard of the section weighs 6.6 kN and its C.G is 3.05 m from A-A.



Sol: $M_a + 135 = 200 + 200$
 $M_a = 265$
 $a = 265 / 135 = 1.969$

$N + M_{G.a} + 2.25 = 200$
 $N = 193.3 \text{ kN (After Solving)}$

Moment, $M + N \times 0.15 = 0$, $M = -29. \text{ kN-m.}$

Shear force & Bending Moment.

Shear force $S - M_{W.a} - 6.6 = 0$

Bending Moment $M - M_{W.a} \times 3.05 - 6.6 \times 3.05 = 0$

$M = 59.6 \text{ kN-m.}$